Q1:

```
import java.util.*;

class Course {
    HashSet<String> students = new HashSet<>();
    ArrayList<String> prerequisites = new ArrayList<>();
    LinkedList<String> waitlist = new LinkedList<>();
    int capacity;

    Course(int capacity) {
        this.capacity = capacity;
    }

    void register(String student) {
        if (students.size() < capacity) {
            students.add(student);
            System.out.println(student + " enrolled");
        } else {
            waitlist.add(student);
            System.out.println(student + " added to waitlist");
        }
    }

    void drop(String student) {
        students.remove(student);

        if (!waitlist.isEmpty()) {
            students.add(waitlist.removeFirst());
        }
    }
}
```

```
public class CourseRegistration {
    public static void main(String[] args) {

        HashMap<String, Course> courses = new HashMap<>();

        Course c1 = new Course(2);

        c1.prerequisites.add("CS101");
        c1.prerequisites.add("MA101");

        courses.put("CS201", c1);

        courses.get("CS201").register("S1");
        courses.get("CS201").register("S2");
        courses.get("CS201").register("S3");

        courses.get("CS201").drop("S1");

        System.out.println("Enrolled: " + courses.get("CS201").students);
        System.out.println("Prerequisites: " + courses.get("CS201").prerequisites);
        System.out.println("Waitlist: " + courses.get("CS201").waitlist);
    }
}
```

Q2)

```
import java.util.*;

class Song {
    String name;
```

```
    Song(String name) {
        this.name = name;
    }

    public String toString() {
        return name;
    }

    public boolean equals(Object obj) {
        Song s = (Song) obj;
        return name.equals(s.name);
    }

    public int hashCode() {
        return name.hashCode();
    }
}

public class MediaPlayer {
    public static void main(String[] args) {

        LinkedList<Song> playlist = new LinkedList<>();

        playlist.add(new Song("Song 1"));
        playlist.add(new Song("Song 2"));
        playlist.add(new Song("Song 3"));

        playlist.addFirst(new Song("Priority Song"));

        HashSet<Song> unique = new HashSet<>();
```

```
        ListIterator<Song> iterator = playlist.listIterator();

        while (iterator.hasNext()) {
            Song song = iterator.next();

            if (!unique.add(song)) {
                iterator.remove();
            }
        }

        System.out.println("Forward:");

        iterator = playlist.listIterator();

        while (iterator.hasNext()) {
            System.out.println(iterator.next());
        }

        System.out.println("Backward:");

        iterator = playlist.listIterator(playlist.size());

        while (iterator.hasPrevious()) {
            System.out.println(iterator.previous());
        }
    }
}
```